EXPERIMENT 13

AIM:

To handle missing, duplicate, and inappropriate data in a given dataset using the Pandas library.

CODE:

```
import numpy as np

import pandas as pd


# Load the dataset

df = pd.read_csv("student_missing_inappropriate.csv")

print(df)


# Identify duplicate records

print(df.duplicated())


# Display dataset information

df.info()


# Remove duplicate records

df.drop_duplicates(inplace=True)

print(df)


# Display number of records

print("Number of records:", len(df))
```

```
# Reset the index
index = np.array(list(range(0, len(df))))
df.set_index(index, inplace=True)
print(index)

# Display the dataset
print(df)

# Replace negative Marks with missing values
df.loc[df["Marks"] < 0, "Marks"] = np.nan
print(df)

# Replace negative Attendance with missing values
df.loc[df["Attendance"] < 0, "Attendance"] = np.nan
print(df)

# Replace negative Fees with missing values
df.loc[df["Fees"] < 0, "Fees"] = np.nan
print(df)

# Display unique departments
print(df["Department"].unique())

# Fill missing Marks using median
df["Marks"].fillna(
```

```
    round(df["Marks"].median()),
    inplace=True
)
print(df)

# Fill missing Attendance using median
df["Attendance"].fillna(
    round(df["Attendance"].median()),
    inplace=True
)
print(df)

# Fill missing Fees using mean
df["Fees"].fillna(
    round(df["Fees"].mean()),
    inplace=True
)
print(df)

# Display the final cleaned dataset
print(df)
```

OUTPUT:

1. Original Dataset

Index	StudentID	Name	Age	Marks	Dept.	Attend.	Fees
0	1	Arun	19	85	CSE	92	50000
1	2	Priya	20	92	CSE	88	52000
2	3	Ravi	19	76	ECE	75	48000
3	4	Divya	20	-5	EEE	91	51000
4	5	Kiran	21	88	CSE	86	53000
5	5	Kiran	21	88	CSE	86	53000
6	7	Meena	20	81	ECE	80	-1000
7	8	Ajay	19	95	EEE	95	50000
8	9	Sneha	20	-10	CSE	89	54000
9	10	Vijay	22	90	CSE	-1	55000

2. Duplicate Check

df.duplicated()
0 False
1 False
2 False
3 False
4 False
5 True
6 False
7 False
8 False
9 False
dtype: bool

The record at index 5 is a duplicate and is removed using df.drop_duplicates(inplace=True).

3. Dataset Information (df.info())

RangeIndex: 10 entries, 0 to 9
Data columns (total 7 columns):
StudentID 10 non-null int64
Name 10 non-null object
Age 10 non-null int64
Marks 10 non-null int64
Department 10 non-null object

Attendance 10 non-null int64
Fees   10 non-null int64
dtypes: int64(5), object(2)
memory usage: 692.0+ bytes

4. Dataset After Removing Duplicate

Index	StudentID	Name	Age	Marks	Dept.	Attend.	Fees
0	1	Arun	19	85	CSE	92	50000
1	2	Priya	20	92	CSE	88	52000
2	3	Ravi	19	76	ECE	75	48000
3	4	Divya	20	-5	EEE	91	51000
4	5	Kiran	21	88	CSE	86	53000
6	7	Meena	20	81	ECE	80	-1000
7	8	Ajay	19	95	EEE	95	50000
8	9	Sneha	20	-10	CSE	89	54000
9	10	Vijay	22	90	CSE	-1	55000

len(df) = 9

5. Replacing Inappropriate Negative Values with NaN

Negative values in Marks, Attendance, and Fees are treated as missing values. Output after applying the three replacements:

Index	StudentID	Name	Age	Marks	Dept.	Attend.	Fees
0	1	Arun	19	85.0	CSE	92.0	50000.0
1	2	Priya	20	92.0	CSE	88.0	52000.0
2	3	Ravi	19	76.0	ECE	75.0	48000.0
3	4	Divya	20	NaN	EEE	91.0	51000.0
4	5	Kiran	21	88.0	CSE	86.0	53000.0
5	7	Meena	20	81.0	ECE	80.0	NaN
6	8	Ajay	19	95.0	EEE	95.0	50000.0
7	9	Sneha	20	NaN	CSE	89.0	54000.0
8	10	Vijay	22	90.0	CSE	NaN	55000.0

6. Unique Departments

array(['CSE', 'ECE', 'EEE'], dtype=object)

7. Missing-Value Imputation

Column	Method Used	Replacement Value
Marks	Median, rounded	88
Attendance	Median, rounded	88
Fees	Mean, rounded	51625

The notebook fills missing Marks and Attendance with their rounded medians, and missing Fees with the rounded mean.

8. Final Cleaned Dataset

Index	StudentID	Name	Age	Marks	Dept.	Attend.	Fees
0	1	Arun	19	85.0	CSE	92.0	50000.0
1	2	Priya	20	92.0	CSE	88.0	52000.0
2	3	Ravi	19	76.0	ECE	75.0	48000.0
3	4	Divya	20	88.0	EEE	91.0	51000.0
4	5	Kiran	21	88.0	CSE	86.0	53000.0
5	7	Meena	20	81.0	ECE	80.0	51625.0
6	8	Ajay	19	95.0	EEE	95.0	50000.0
7	9	Sneha	20	88.0	CSE	89.0	54000.0
8	10	Vijay	22	90.0	CSE	88.0	55000.0

RESULT:

Duplicate records and inappropriate values were successfully identified and handled, and the dataset was cleaned using Pandas.